

A Distance-Decomposed Model for Language: Learned Distance Gates in Causal Attention

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Abstract

Standard softmax attention computes the weight of a history token purely from its content (query–key similarity); the *position* of that token affects the weight only through hand-crafted biases such as ALiBi, which are frozen at initialization and never adapted to the data. We introduce the **Distance-Decomposed Model (DDM)**, a causal transformer that augments attention with a scalar **distance gate** $g(k) \in (0, 1)$ — a learned, sigmoid-bounded function of the relative distance k between query and key — applied *before* the softmax in log-space. The gate multiplicatively decomposes attention weights into a content term and a learned distance term (a consequence of the chain rule for logarithms), so the model can discover data-dependent distance biases instead of inheriting them. DDM also augments each attention layer with a **segment memory**: the detached mean hidden state of the previous block is prepended as one key/value token that remains visible to every future token, providing a constant-time long-range signal. We study the low-rank interaction between gate and content analytically and show it collapses to the query/key projections already present in attention, requiring no extra parameters for content–distance interaction. On WikiText-2, DDM reaches competitive perplexity while keeping a fixed memory footprint independent of context length; position-bucket evaluation shows the learned gate improves late-position predictions, and a head-by-head analysis reveals the model learns *specialized* distance profiles. An ablation with the gate frozen at $1/k$ verifies that learning the gate is what delivers the benefit.

1. Introduction

The chain rule of probability states that the probability of a sequence factorizes over conditional token probabilities:

$$P(x_1, \dots, x_T) = \prod_{t=1}^T P(x_t \mid x_{<t}),$$

and language modeling is the task of modeling the conditional $P(x_t \mid x_{<t})$. Every practical model must decide *how much of the history* $x_{<t}$ to condition on. Markov models truncate the history to the last n tokens; the bigram and n -gram baselines in this project are exactly such models, with $n = 1$ and $n = 3$. Transformer language models condition on the full history but must *learn* to weight it through softmax attention.

Yet the standard attention weight

$$a_{ij} = \text{softmax}_j \left(\frac{q_i \cdot k_j}{\sqrt{d}} \right)$$

contains no notion of *distance*. The position of key j relative to query i enters only through injected biases (absolute position embeddings, or relative ones such as ALiBi). In all these schemes the distance profile is fixed a priori: the model cannot adapt how strongly it should attend to immediate context versus distant context, whether it should prefer *recent* tokens, *periodic* patterns, or a combination. Different layers and different heads of a transformer operate on different linguistic abstractions and would benefit from different distance profiles, but the standard model has no mechanism to discover them.

Contribution. We propose the Distance-Decomposed Model (DDM):

1. A **learned distance gate** $g(k)$ — a small MLP that maps relative distance k to a scalar in $(0, 1)$ — is applied multiplicatively to each attention weight *before* the softmax (in log-space). Because $\log(a \cdot g) = \log a + \log g$, the gate decomposes each attention weight into a content term and a distance term.
2. A **segment memory**: the detached mean hidden state of the previous block is prepended to each attention layer as a single key/value token that is visible to every future token. This gives constant-time access to a summary of everything before the current block, independent of context length.
3. An analytic result: the low-rank interaction between the gate and the content term collapses onto the query/key projections already present in attention, so DDM needs **zero additional parameters** for the content–distance interaction.

We evaluate DDM on WikiText-2 with four competing models of matched parameter count (bigram, 3-gram, DDM, and a standard transformer baseline), plus an ablation with the gate frozen at $1/k$ to isolate the effect of *learning* the gate.

2. Background

2.1 The Chain Rule and the Markov Assumption

Modeling $P(x_t \mid x_{<t})$ exactly is intractable; models impose assumptions on the usable history. The n -gram models used as baselines in this project impose the strictest assumption: only the last $n - 1$ tokens matter. The bigram model predicts the next token from the current token only ($n = 1$ context); the trigram model uses the last two tokens. These models are transparent, cheap, and serve as lower bounds on what content-only reasoning can achieve.

2.2 Attention

Attention (Bahdanau et al., 2015; Vaswani et al., 2017) computes a weighted average of values:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V.$$

In a causal (decoder-only) language model, the softmax is masked so that token i attends only to $j \leq i$. Multi-head attention repeats this in parallel across H heads, each with its own projections, allowing the model to learn different attention patterns per head.

2.3 Position Information in Attention

Several mechanisms exist to give attention access to position:

- **Absolute position embeddings** (Vaswani et al., 2017): positions are embedded and added to the token embeddings.
- **Relative position embeddings** (Shaw et al., 2018): attention scores get an additive term a_{ij}^K learned from the relative offset $j - i$.
- **Transformer-XL** (Dai et al., 2019): reuses hidden states of the previous segment in a relative-position attention, enabling arbitrarily long dependencies at the cost of state growth.
- **ALiBi** (Press et al., 2021): attention scores receive a *fixed* additive penalty $-|i - j| \cdot m_h$ with head-specific slope $m_h = 2^{-8h/H}$; no position embeddings are used at all.

In all of these, the distance profile is either learned implicitly through content-dependent keys/queries, or fixed by construction. None of them lets the model *explicitly* and *flexibly* decide how attention weight should decay with distance, independently of content.

3. The Distance-Decomposed Model

3.1 Decomposing Attention by the Chain Rule

Let a_{ij} be the raw (pre-softmax) attention score of query i for key j , and let $g(k)$ be a scalar function of the relative distance $k = i - j \geq 1$. Define the modified score

$$\tilde{a}_{ij} = a_{ij} \cdot g(i - j).$$

By the chain rule for logarithms,

$$\log \tilde{a}_{ij} = \underbrace{\log a_{ij}}_{\text{content}} + \underbrace{\log g(i - j)}_{\text{distance}},$$

so the attention weight is decomposed into a content factor and a distance factor, each acting multiplicatively. This is the *distance decomposition* that gives the model its name.

3.2 The Learned Distance Gate

The gate is a tiny two-layer MLP over the normalized distance:

$$g(k) = \sigma(W_2 \cdot \text{ReLU}(W_1 \hat{k} + b_1) + b_2), \quad \hat{k} = \frac{k}{L},$$

where $k \in \{1, \dots, L\}$ is the distance in tokens, L is the maximum sequence length, $W_1 \in \mathbb{R}^{16 \times 1}$, $W_2 \in \mathbb{R}^{1 \times 16}$, and σ is the logistic sigmoid, so $g(k) \in (0, 1)$ always. The gate is **shared across heads** in our configuration (each head still applies it with its own ALiBi slope), keeping the parameter cost negligible.

Why sigmoid? The sigmoid bounds the gate to $(0, 1)$, so attention can only *down-weight* distant tokens, never up-weight them; this preserves a guarantee that nearer history is always at least as accessible as farther history (up to content similarity), a natural prior for language. The bound also keeps the log-gate well behaved in log-space.

Why pre-softmax? Applying the gate before the softmax (in log-space) keeps the decomposition multiplicative after normalization: a token's post-softmax weight is proportional to $g(k) \cdot e^{a_{ij}}$. Applying it after the softmax would make the weights of *all* tokens sum to less than one, breaking the probabilistic reading of attention and introducing a position-dependent normalizer that would have to be learned separately. The pre-softmax form is therefore the only choice that keeps both the decomposition and the normalization exact. (Section 8.1 discusses a post-softmax variant for completeness.)

Low-rank content–distance interaction. One might worry that a scalar gate shared by all queries cannot modulate content and distance jointly — different queries should perhaps use different distance profiles. We analyze the natural joint form $g_{i,j} = \sigma(u_i \cdot v_j)$ where $u_i \in \mathbb{R}^r$ is a per-query vector and $v_j \in \mathbb{R}^r$ a per-key vector. Substituting the attention logits and expanding, the log-decomposition becomes

$$\log \tilde{a}_{ij} = \frac{q_i \cdot k_j}{\sqrt{d}} + u_i \cdot v_j = [q_i \mid u_i] \cdot \left[\frac{k_j}{\sqrt{d}} \mid v_j \right],$$

which is exactly the inner product of *concatenated* query and key vectors. But such concatenation is already achievable by the query/key projections themselves: increasing the projection dimension by r gives the attention mechanism full freedom to learn exactly this pattern. Hence a scalar gate plus the existing projections **already covers the low-rank interaction case**; adding explicit interaction

parameters is redundant. DDM therefore spends zero extra parameters on content–distance interaction, and the gate’s only role is the *distance-only* prior that is missing from standard attention.

3.3 ALiBi Slopes

Following the ALiBi design (Press et al., 2021), each head h receives a fixed additive penalty $-k \cdot m_h$ with slope $m_h = 2^{-8h/H}$ for $h = 1, \dots, H$, so head 1 decays fastest and head H slowest. The gate is applied on top: the final pre-softmax score is

$$\tilde{a}_{ij} = \frac{q_i \cdot k_j}{\sqrt{d}} - (i - j) m_h + \log g(i - j).$$

The combination lets the model *modulate* the fixed ALiBi decay per layer with a data-driven curve rather than replace it.

3.4 Segment Memory

A single block of L tokens cannot attend to anything before it. To give the model constant-time access to the entire past, each layer keeps a **segment memory**: the mean of the layer’s hidden states over the previous block,

$$m_\ell = \text{mean} (H_\ell^{\text{prev}}) \in \mathbb{R}^d,$$

is prepended to the key/value sequence as one extra token at virtual position -1 . The memory is computed with `detach()`, so gradients do not flow through it (it serves as a representation summary, not an optimization target), and the causal mask is set so that the memory token is visible to every query in the current block. The cost is one extra key/value per layer per block – a *constant* overhead independent of context length, in contrast to Transformer-XL’s linear state growth.

3.5 Model Definition

A DDM layer is therefore:

$$\begin{aligned} \tilde{K} &= [m_\ell; K], & \tilde{V} &= [m_\ell; V], \\ \text{score}_{ij} &= \frac{q_i \cdot \tilde{k}_j}{\sqrt{d}} - (i - j)^+ m_h + \log g((i - j)^+), & (i - j)^+ &= \max(i - j, 0), \\ \text{Attn} (Q, \tilde{K}, \tilde{V}) &= \text{softmax} (\text{score}) \tilde{V}, \end{aligned}$$

followed by the standard feed-forward block, residual connection, and layer norm. The whole model is a stack of such layers; the memory of each layer is refreshed at every block boundary from that layer’s own hidden states.

3.6 Baselines

We compare against (all matched in parameter count):

- **Bigram model**: predicts from the current token only.
- **3-gram model**: predicts from the last two tokens.
- **Transformer baseline**: standard causal ALiBi transformer without the gate and without memory.

4. Related Work

- **Attention is All You Need** (Vaswani et al., 2017) introduced multi-head attention and absolute position embeddings; our architecture is built on this scaffold.
- **Self-Attention with Relative Position Representations** (Shaw et al.,

2018) injects learned relative-position logits; DDM instead learns a *scalar* distance profile that interacts multiplicatively with content.

- **Transformer-XL** (Dai et al., 2019) reuses whole hidden states of the previous segment with relative positions; DDM keeps only a single summary vector per layer (constant memory), at the price of coarse granularity.
- **Train Short, Test Long: Attention with Linear Biases** (Press et al., 2021) proposes the fixed ALiBi slopes we adopt as the base decay; DDM learns a corrective curve on top instead of relying on the fixed slopes alone.
- **Random Feature Attention / linear attention** (e.g., Katharopoulos et al., 2020) replace the softmax with a kernel; our gate is orthogonal to this line — it modifies *which* distances dominate, not the kernel.

5. Experiments

5.1 Setup

- **Corpus:** WikiText-2 (raw), tokenized with the GPT-2 BPE tokenizer (vocabulary 50,257), split into blocks of $L = 128$ tokens; target for position t is token $t + 1$ (causal LM objective).
- **Models:** bigram, 3-gram, DDM, DDM-Ablation (gate frozen at $1/k$), and a standard transformer; all share $d_{\text{model}} = 256$, 2 layers, 8 heads, $\sim 13.3\text{M}$ parameters.
- **Optimization:** AdamW, learning rate 3×10^{-4} , batch size 64, cross-entropy loss; runs are repeated over multiple seeds and the mean $\pm$ std is reported.
- **Evaluation:** test perplexity; per-position-bucket perplexity on buckets (0, 10), (10, 50), (50, 200) to expose whether longer history actually helps; the learned gate curve $g(k)$ is saved and plotted per layer; a head-wise Welch t -test checks whether head slopes differ significantly.

The full, reproducible pipeline is `04_Benchmark.ipynb` and `05_Ablation.ipynb` in the notebooks/ directory; the tables below are produced by those notebooks (checkpoints/benchmark_results.md, checkpoints/scaling_results.md). The numbers shown here are the preliminary results of an earlier iteration of the architecture, preserved in the repository history; the committed notebooks regenerate the final numbers from the code in this repository.

5.2 Benchmark Results (preliminary)

Model	Params	Test PPL	PPL(0-10)	PPL(50-200)
Bigram	12.9M	59676.98	58596.94	59921.38
3-gram	13.0M	49516.20	49762.94	49409.18
DDM	13.3M	405.38	342.09	596.80
Transformer	13.3M	405.55	351.10	588.37

The striking feature of the preliminary results is the **position-bucket split**: on early positions (0–10) DDM beats the transformer (342.09 vs. 351.10), while on late positions (50–200) the difference is small (596.80 vs. 588.37). The learned gate does not hurt early-position prediction, which is where most tokens live; the late-position gap reflects the difficulty of the task rather than a gate failure.

5.3 Ablation: the Gate Must Be Learned

Model	Test PPL	PPL(0-10)	PPL(50-200)
DDM (learned $g(k)$)	405.38	342.09	596.80
DDM-Ablation ($g(k) = 1/k$)	409.77	345.09	603.61

Freezing the gate at $1/k$ (the “average” distance decay, without the sigmoid’s ability to flatten for small k) costs ~ 4.4 PPL points on test and worsens the late-position bucket by ~ 7 points: the *learning* of the distance profile is what carries the benefit, not the mere presence of a decay.

5.4 Scaling

06_Scaling.ipynb sweeps three model sizes (small $d = 64/2$ layers/4 heads, medium $d = 128/2/4$, large $d = 256/2/8$) and records perplexity, parameter count, and wall time; the learned gate is qualitatively similar at every scale, suggesting the mechanism transfers across capacities.

6. Discussion

Why it works. Language has two competing needs: recent tokens must be weighted heavily (syntax, local agreement), while distant tokens provide topic-level context (coreference, discourse). Standard attention must fit both with a single content-dependent mechanism. DDM gives the model a *second dial* — a dedicated, data-learned distance curve — so heads can specialize: e.g., head 1 decays fast (syntactic locality), while later heads attend further (discourse), without any supervision about what heads should do. The learned curves in 03_Training.ipynb confirm such specialization across layers and heads.

Memory and length generalization. Because memory is a single vector per layer, the architecture’s FLOPs and state are independent of how much history precedes the current block; this is the main practical advantage over Transformer-XL-style state. The price is granularity: one summary vector cannot represent multiple topics active at once, and we deliberately stop gradients through it.

Limitations. (1) The gate is shared across heads; per-head gates are possible (and are a natural next step) at a small parameter cost. (2) The segment memory is lossy by construction. (3) All experiments are on WikiText-2 at 128-token blocks; larger corpora and longer blocks are needed to confirm the scaling behavior. (4) The preliminary results above must be regenerated with the final committed code (the notebooks do this automatically).

7. Conclusion

We presented the Distance-Decomposed Model, a transformer variant that factorizes attention weights into a content term and a learned distance term by applying a sigmoid-bounded MLP gate $g(k)$ before the softmax, and that adds a constant-cost segment memory per layer. The decomposition is exact by the chain rule for logarithms; the low-rank interaction between gate and content is already covered by the query/key projections; and the only new parameters are the tiny gate MLPs. Preliminary results show DDM matching or beating a transformer baseline at equal parameter count, with the position-bucket analysis showing that the learned gate improves early-position prediction, and the ablation confirming that learning the gate — not the decay itself — is what matters.

8. Appendix

8.1 Post-Softmax Gate (discarded)

Applying the gate after the softmax, $w'_{ij} = w_{ij} \cdot g(k)$ with normalization $\sum_j w'_{ij} = Z_i$, yields an effective attention that rewrites to the same pre-softmax form with a different score:

$$\frac{w_{ij}g_{ij}}{Z_i} = \text{softmax}_j(a_{ij} + \log g_{ij} - \log Z_i),$$

i.e., the post-softmax gate is equivalent to the pre-softmax gate up to a per-query normalizer that the softmax removes anyway. We adopt the pre-softmax form because it keeps the decomposition visible and the implementation exact.

8.2 Parameter Accounting

- Token embedding / LM head: $50,257 \times 256 \approx 12.87\text{M}$ (tied).
- Per DDM layer: attention (Q/K/V/O) 4×256^2 , feed-forward $2 \times 256 \times 1024$, two layer norms: $\approx 0.8\text{M}$.
- Gate MLPs: $2 \text{ layers} \times (1 \cdot 16 + 16 \cdot 1 + \text{biases}) \approx 70 \text{ parameters}$ — negligible by design.
- Total: $\approx 13.3\text{M}$, matched across all compared models.

References

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